ACI 318-19 Example 002

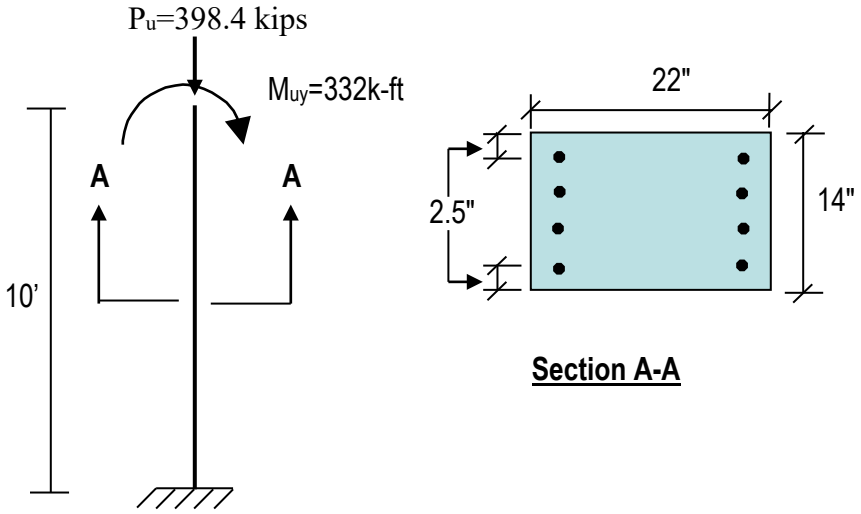
P-M Interaction Check for Compression-Controlled Rectangular Column

Example Description

The Demand/Capacity ratio for a given axial loading and moment is tested in this example.

A reinforced concrete column is subjected factored axial load P_u = 398.4 k and moments M_{uy} = 332 k-ft. This column is reinforced with 4 #9 bars. The total area of reinforcement is 8.00 in^2. The design capacity ratio is checked by hand calculations and result is compared.

Geometry, Properties and Loading



Section A-A

Material Properties		Section Properties		Design Properties	
E =	3600 k/in^2	b =	14 in	f'_c =	4 k/in^2
v =	0.2	d =	19.5 in	f_y =	60 k/in^2
G =	1500 k/in^2				

TECHNICAL FEATURES TESTED

- Tied Reinforced Concrete Column Demand/Capacity Ratio

RESULTS COMPARISON

Independent results are hand calculated and compared.

Output Parameter	ETABS	Independent	Percent Difference
Column Demand/Capacity Ratio	1.000	1.00	0.00%

COMPUTER FILE: ACI 318-19 Ex002

CONCLUSION

The computed results show an exact match with the independent results.

HAND CALCULATION

COLUMN STRENGTH UNDER COMPRESSION CONTROL

$f'_c = 4$ ksi $f_y = 60$ ksi
b = 14 inch d = 19.5 inch
$P_u = 398.4$ kips $M_u = 332$ k-ft

1) Because $e = 10$ inch $< (2/3)d = 13$ inch., assume compression failure. This assumption will be checked later. Calculate the distance to the neutral axis fro a balanced condition, c_b:

Position of neutral axis at balance condition:

$$c_b = \frac{87}{87 + f_y} d_t = \frac{87}{87 + 60}(19.5) = 11.54 \text{ inch}$$

2) From the equation of equilibrium:

$$P_n = C_c + C_s - T$$

where

$$C_c = 0.85 f'_c ab = 0.85 \bullet 4 \bullet 14a = 47.6a$$

$$C_s = A'_s \left(f_y - 0.85 f'_c \right) = 4\left(60 - 0.85 \bullet 4\right) = 226.4 \text{ kips}$$

Assume compression steels yields, (this assumption will be checked later).

$$T = A_s f_s = 4 f_s \left(f_s < f_y \right)$$

$$P_n = 47.6a + 226.4 - 4 f_s \qquad \text{(Eqn. 1)}$$

3) Taking moments about A_s:

$$P_n = \frac{1}{e'}\left[C_c \left(d - \frac{a}{2} \right) + C_s \left(d - d' \right) \right]$$

The plastic centroid is at the center of the section and $d'' = 8.5$ inch

$$e' = e + d'' = 10 + 8.5 = 18.5 \text{ inch.}$$

$$P_n = \frac{1}{18.5}\left[47.6a\left(19.5 - \frac{a}{2} \right) + 226.4\left(19.5 - 2.5\right) \right]$$

$$P_n = 50.17a - 1.29a^2 + 208 \qquad \text{(Eqn. 2)}$$

4) Assume c = 13.45 inch, which exceed c_b (11.54 inch).

$$a = 0.85 \bullet 13.45 = 11.43 \text{ inch}$$

Substitute in Eqn. 2:

$$P_n = 50.17 \bullet 11.43 - 1.29 \bullet (11.43)^2 + 208 = 612.9 \text{ kips}$$

5) Calculate f_s from the strain diagram when c = 13.45 inch.

$$f_s = \left(\frac{19.5 - 13.45}{13.45}\right)87 = 39.13 \text{ ksi}$$

$$\varepsilon_s = \varepsilon_t = f_s / E_s = 0.00135$$

6) Substitute a = 13.45 inch and f_s = 39.13 ksi in Eqn. 1 to calculate P_{n2}:

$$P_{n2} = 47.6(11.43) + 226.4 - 4(39.13) = 613.9 \text{ kips}$$

Which is very close to the calculated P_{n2} of 612.9 kips (less than 1% difference)

$$M_n = P_n e = 612.9\left(\frac{10}{12}\right) = 510.8 \text{ kips-ft}$$

7) Check if compression steels yield. From strain diagram,

$$\varepsilon_s' = \left(\frac{13.45 - 2.5}{13.45}\right)(0.003) = 0.00244 > \varepsilon_y = 0.00207 \text{ ksi}$$

Compression steels yields, as assumed.

8) Calculate ϕ,

$d_t = d = 19.5$ inch, $\quad c = 13.45$ inch

ε_t (at the tension reinforcement level) $= 0.003\left(\frac{19.45 - 13.45}{13.45}\right) = 0.00135$

Since $\varepsilon_t < 0.002$, then $\phi = 0.65$

$$\phi P_n = 0.65(612.9) = 398.4 \text{ kips}$$

$$\phi M_n = 0.65(510.8) = 332 \text{ k-ft.}$$

9) Calculate the D/C ratio.

$$\text{D/C} = \frac{P_u}{\phi P_n} = \frac{398.4}{398.4} = 1.00$$

$$\text{D/C} = \frac{M_u}{\phi M_n} = \frac{332}{332} = 1.00$$